THE COMPARISON MAP FROM THE PULLED BACK OBJECT TENSORED WITH THE SHRIEK OF THE STRUCTURE SHEAF TO THE SHRIEK IS WELL DEFINED AND ASSOCIATIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0B6T (sources/stacks-duality/pinned-original-source.tex line 4427).

Allowed standard prerequisites (editorial).

- The upper shriek functor through a compactification and its well definedness. Same chapter, Lemma labelled lemma-shriek-well-defined, cited at proof lines 4470-4474 and repeatedly as the model argument, and Lemma labelled lemma-twisted-inverse-image for the right adjoint. Supplies the description f^! = j^* o a-bar and the independence argument to be imitated.
- The comparison map between pullback and the right adjoint for a proper morphism, and its transitivity. Same chapter, equation labelled equation-compare-with-pullback, cited at proof lines 4458-4461 and 4469-4472, and Lemma labelled lemma-transitivity-compare-with-pullback, cited at proof lines 4461 and 4520-4524 and 4552-4556. Produces the maps whose restrictions define mu and proves the commutativity diagrams.
- Base change maps for the right adjoints and their restriction to the open. Same chapter, equation labelled equation-sheafy, cited at proof lines 4516-4520, Lemma labelled lemma-compare-on-open, cited at proof lines 4526-4530, and Lemma labelled lemma-restriction-compare-with-pullback, cited at proof lines 4534-4538. Shows that sigma and tau restrict to isomorphisms and identifies the restrictions with the identity.
- Composition of the shriek functors. Same chapter, Lemma labelled lemma-upper-shriek-composition, cited at proof lines 4544-4550. Supplies the construction of (g o f)^! -> f^! o g^! used for the hexagon.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 0A9S: sources/stacks-duality/pinned-original-source.tex, source lines 2079–2079. Same-chapter pointer into the author environment equation at source line 2079; invoked at proof line 4458 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B6R: sources/stacks-duality/pinned-original-source.tex, source lines 2189–2189. Same-chapter pointer into the author environment lemma at source line 2189; invoked at proof line 4459 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AA0: sources/stacks-duality/pinned-original-source.tex, source lines 3986–3986. Same-chapter pointer into the author environment lemma at source line 3986; invoked at proof line 4474 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9E: sources/stacks-duality/pinned-original-source.tex, source lines 653–653. Same-chapter pointer into the author environment lemma at source line 653; invoked at proof line 4481 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B6Q: sources/stacks-duality/pinned-original-source.tex, source lines 2171–2171. Same-chapter pointer into the author environment lemma at source line 2171; invoked at proof line 4512 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 0A9L: sources/stacks-duality/pinned-original-source.tex, source lines 988–988. Same-chapter pointer into the author environment equation at source line 988; invoked at proof line 4515 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B6P: sources/stacks-duality/pinned-original-source.tex, source lines 2119–2119. Same-chapter pointer into the author environment lemma at source line 2119; invoked at proof line 4523 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ATX: sources/stacks-duality/pinned-original-source.tex, source lines 4161–4161. Same-chapter pointer into the author environment lemma at source line 4161; invoked at proof line 4553 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0F42 let $f : X \to Y$ be a morphism of FTS_S. There are canonical maps*

$$\mu_{f,K} : Lf^*K \otimes^{\mathbf{L}}_{\mathcal{O}_X} f^!\mathcal{O}_Y \longrightarrow f^!K$$

functorial in K in $D^+_{QCoh}(\mathcal{O}_Y)$. If $g : Y \to Z$ is another morphism of FTS_S, then the diagram

$$\begin{array}{ccccc} Lf^*(Lg^*K \otimes^{\mathbf{L}}_{\mathcal{O}_Y} g^!\mathcal{O}_Z) \otimes^{\mathbf{L}}_{\mathcal{O}_X} f^!\mathcal{O}_Y & \xrightarrow[\mu_f]{} & f^!(Lg^*K \otimes^{\mathbf{L}}_{\mathcal{O}_Y} g^!\mathcal{O}_Z) & \xrightarrow[f^!\mu_g]{} & f^!g^!K \\ \| & & & & \| \\ Lf^*Lg^*K \otimes^{\mathbf{L}}_{\mathcal{O}_X} Lf^*g^!\mathcal{O}_Z \otimes^{\mathbf{L}}_{\mathcal{O}_X} f^!\mathcal{O}_Y & \xrightarrow{\mu_f} & Lf^*Lg^*K \otimes^{\mathbf{L}}_{\mathcal{O}_X} f^!g^!\mathcal{O}_Z & \xrightarrow{\mu_{g\circ f}} & f^!g^!K \end{array}$$

commutes for all $K \in D^+_{QCoh}(\mathcal{O}_Z)$.

Proof. If f is proper, then $f^! = a$ and we can use (0A9S) and if g is also proper, then Lemma 0B6R proves the commutativity of the diagram (in greater generality).

Let us define the map $\mu_{f,K}$. Choose a compactification $j : X \to \overline{X}$ of X over Y. Since $f^!$ is defined as $j^* \circ \overline{a}$ we obtain $\mu_{f,K}$ as the restriction of the map (0A9S)

$$L\overline{f}^*K \otimes^{\mathbf{L}}_{\mathcal{O}_{\overline{X}}} \overline{a}(\mathcal{O}_Y) \longrightarrow \overline{a}(K)$$

to X. To see this is independent of the choice of the compactification we argue as in the proof of Lemma 0AA0. We urge the reader to read the proof of that lemma first.

Assume given a morphism $g : \overline{X}_1 \to \overline{X}_2$ between compactifications $j_i : X \to \overline{X}_i$ over Y such that $g^{-1}(j_2(X)) = j_1(X)$. Denote $\overline{c}$ the right adjoint for pushforward of Lemma 0A9E for the morphism g. The maps

$$L\overline{f}_1^*K \otimes^{\mathbf{L}}_{\mathcal{O}_{\overline{X}}} \overline{a}_1(\mathcal{O}_Y) \longrightarrow \overline{a}_1(K) \quad \text{and} \quad L\overline{f}_2^*K \otimes^{\mathbf{L}}_{\mathcal{O}_{\overline{X}}} \overline{a}_2(\mathcal{O}_Y) \longrightarrow \overline{a}_2(K)$$

fit into the commutative diagram

$$\begin{array}{ccccc} Lg^*(L\overline{f}_2^*K \otimes^{\mathbf{L}} \overline{a}_2(\mathcal{O}_Y)) \otimes^{\mathbf{L}} \overline{c}(\mathcal{O}_{\overline{X}_2}) & \xrightarrow[\sigma]{} & \overline{c}(L\overline{f}_2^*K \otimes^{\mathbf{L}} \overline{a}_2(\mathcal{O}_Y)) & \longrightarrow & \overline{c}(\overline{a}_2(K)) \\ \| & & & & \| \\ L\overline{f}_1^*K \otimes^{\mathbf{L}} Lg^*\overline{a}_2(\mathcal{O}_Y) \otimes^{\mathbf{L}} \overline{c}(\mathcal{O}_{\overline{X}_2}) & \xrightarrow{1\otimes\tau} & L\overline{f}_1^*K \otimes^{\mathbf{L}} \overline{a}_1(\mathcal{O}_Y) & \longrightarrow & \overline{a}_1(K) \end{array}$$

by Lemma 0B6R. By Lemma 0B6Q the maps σ and τ restrict to an isomorphism over X. In fact, we can say more. Recall that in the proof of Lemma 0AA0 we used the map (0A9L) $\gamma : j_1^* \circ \overline{c} \to j_2^*$ to construct our isomorphism $\alpha_g : j_1^* \circ \overline{a}_1 \to j_2^* \circ \overline{a}_2$. Pulling back to map σ by j_1 we obtain the identity map on $j_2^*\left(L\overline{f}_2^*K \otimes^{\mathbf{L}} \overline{a}_2(\mathcal{O}_Y)\right)$ if we identify $j_1^*\overline{c}(\mathcal{O}_{\overline{X}_2})$ with $\mathcal{O}_X$ via $j_1^* \circ \overline{c} \to j_2^*$, see Lemma 0B6P. Similarly, the map $\tau : Lg^*\overline{a}_2(\mathcal{O}_Y) \otimes^{\mathbf{L}} \overline{c}(\mathcal{O}_{\overline{X}_2}) \to \overline{a}_1(\mathcal{O}_Y) = \overline{c}(\overline{a}_2(\mathcal{O}_Y))$ pulls back to the identity map on $j_2^*\overline{a}_2(\mathcal{O}_Y)$. We conclude that pulling back by j_1 and applying γ wherever we can we obtain a commutative diagram

$$\begin{array}{ccccc} j_2^*\left(L\overline{f}_2^*K \otimes^{\mathbf{L}} \overline{a}_2(\mathcal{O}_Y)\right) & & \longrightarrow & & j_2^*\overline{a}_2(K) \\ \downarrow & & & \nwarrow^{\alpha_g} & \\ j_1^*L\overline{f}_1^*K \otimes^{\mathbf{L}} j_2^*\overline{a}_2(\mathcal{O}_Y) & \xleftarrow{1\otimes\alpha_g} & j_1^*(L\overline{f}_1^*K \otimes^{\mathbf{L}} \overline{a}_1(\mathcal{O}_Y)) & \longrightarrow & j_1^*\overline{a}_1(K) \end{array}$$

The commutativity of this diagram exactly tells us that the map $\mu_{f,K}$ constructed using the compactification $\overline{X}_1$ is the same as the map $\mu_{f,K}$ constructed using the compactification $\overline{X}_2$ via the identification α_g used in the proof of Lemma 0AA0. Some categorical arguments exactly as in the proof of Lemma 0AA0 now show that $\mu_{f,K}$ is well defined (small detail omitted).

Having said this, the commutativity of the diagram in the statement of our lemma follows from the construction of the isomorphism $(g \circ f)^! \to f^! \circ g^!$ (first part of the proof of Lemma 0ATX using $\overline{X} \to \overline{Y} \to Z$) and the result of Lemma 0B6R for $\overline{X} \to \overline{Y} \to Z$. $\square$